CHAPTER 16 · DERIVATIONS

DIRECT CURRENT CIRCUITS

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derivations the STPM 2026 syllabus asks you to derive, deduce or show

1 CONVERTING A GALVANOMETER: SHUNT AND MULTIPLIER

SYLLABUS 16.4

RESULT $R_s = I_g r_g/(I - I_g)$; $R_m = \dfrac{V}{I_g} - r_g$

1. A galvanometer (milliammeter) of resistance r_g gives full-scale deflection at current I_g.
2. **Ammeter** (range I): connect a shunt R_s of very low resistance in parallel. The p.d. is the same: $V_{shunt} = V_{meter}$, so $(I - I_g)R_s = I_g r_g$ and $\mathbf{R_s = I_g r_g/(I - I_g)}$.
3. **Voltmeter** (range V): connect a multiplier R_m of very high resistance in series. The current is the same: $V = V_{meter} + V_{multiplier} = I_g(r_g + R_m)$, so $\mathbf{R_m = \dfrac{V}{I_g} - r_g}$.

An ammeter must have a very low resistance and a voltmeter a very high resistance, so that they hardly change the circuit.

2 POTENTIOMETER CALCULATIONS

SYLLABUS 16.4

RESULT $V \propto l$

1. **Driver cell only** across the wire (no other resistance): the p.d. across a length l is proportional to l, so $\dfrac{E_1}{E_2} = \dfrac{l_1}{l_2}$.
2. **Driver cell with a resistance** (internal resistance r or a series resistor R): first find the current in the upper (driver) circuit, $I = \dfrac{E}{R_{wire} + R + r}$.
3. Then the p.d. per unit length of wire is $k = \dfrac{IR_{wire}}{L}$, and at balance the e.m.f. of the cell in the lower circuit is $E_x = k\,l$.

At balance no current flows through the galvanometer, so the potentiometer measures the e.m.f., not the terminal p.d.